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SLA: Scaling Spatial-Language-Action Model via Continual Pre-training

Implicit 3D alignment + continual PT: 2D VLA → spatial robot foundation model

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TL;DR

Train-time 3D loss, no inference cost; tops 4 benchmarks + real robot

SLA starts from π0.5 and performs **continual pre-training** on ~1.7M heterogeneous robot trajectories: during training, multi-layer VLA visual tokens are cosine-aligned with **frozen OmniVGGT 3D features** (gated fusion + state dropout), with alignment continued in post-training; **at inference the architecture is identical to π0.5, with no depth or camera calibration needed.**

87.4%

LIBERO-Plus overall SR

π0.5： 84.4% （+3.0）

72.3%

RoboTwin 2.0 Clean2Random

π0.5： 46.0% （+26.3）

33.9%

RoboDojo-Real success rate

π0.5： 12.8% （+21.1）

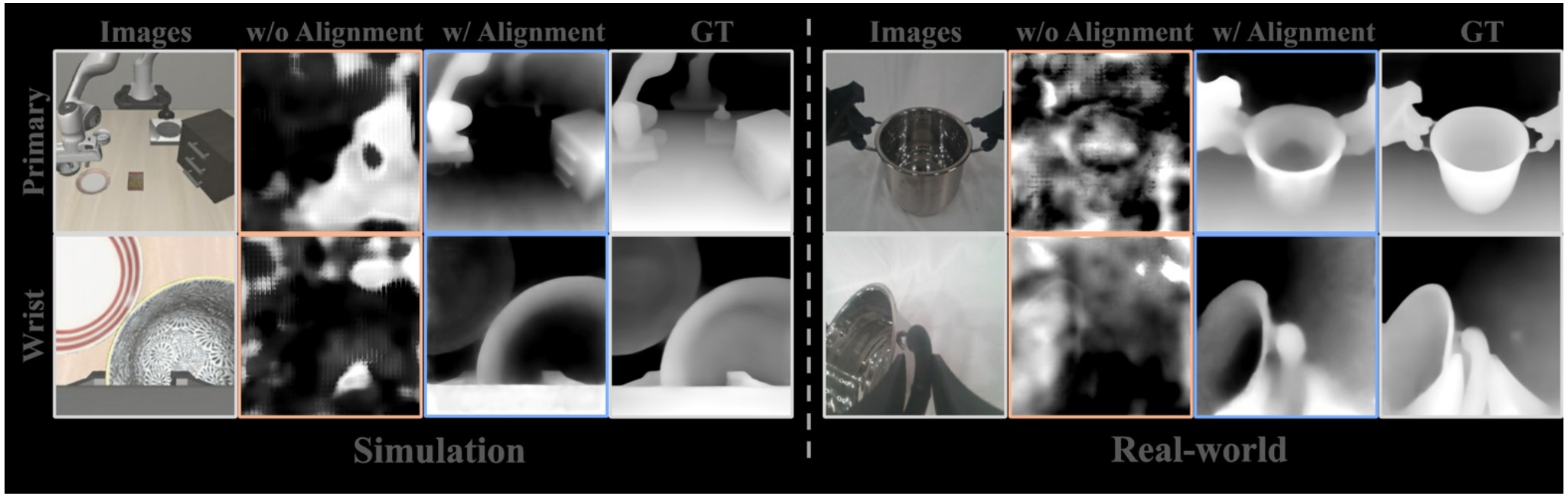
+24.0

Avg., 8 real-robot conds (pp)

83.8% vs π0.5 59.8%

Problem & Motivation

Visual tokens of 2D VLAs contain almost no 3D structure



2D pre-training lacks 3D

Visual representations are inherited from 2D image-text VLMs; spatial supervision is often added only for downstream tasks.

Explicit 3D input scales poorly

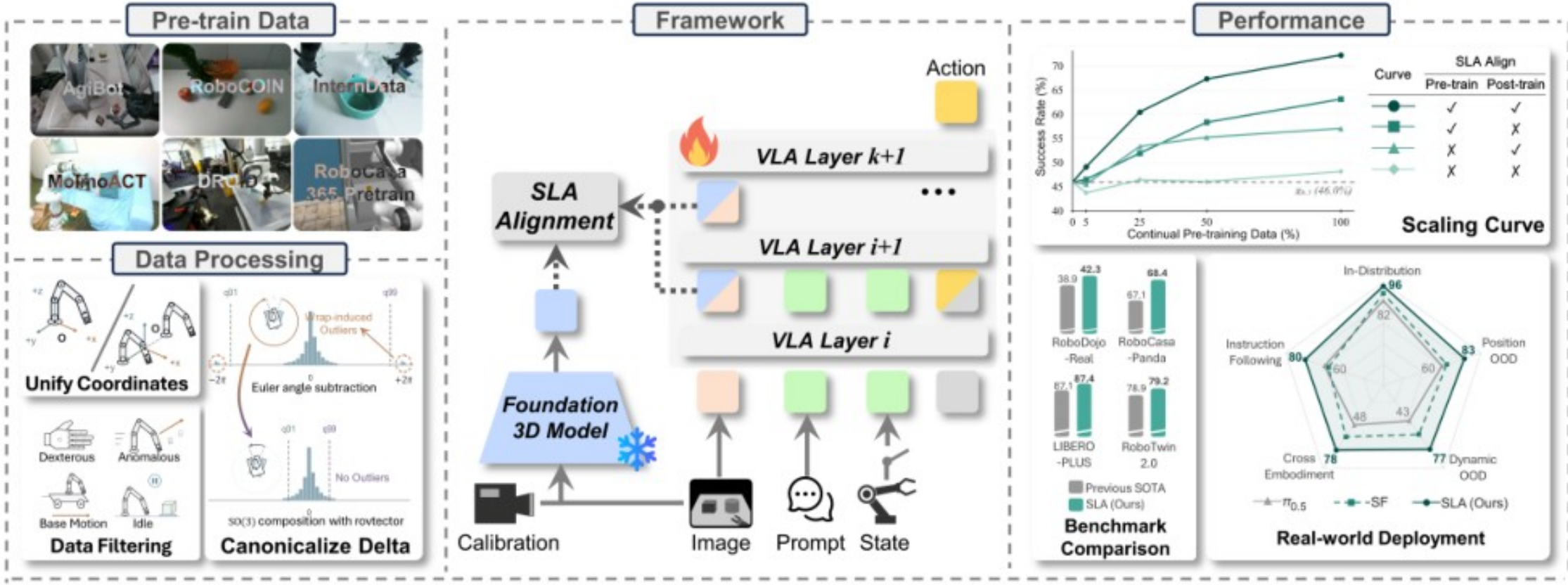
Noisy depth/point clouds, varied camera types and calibration, many public datasets lack depth, and the architecture must change.

Post-train-only align fails

Spatial Forcing aligns implicitly only in post-training; spatial ability is a per-task add-on, not built into the foundation model.

Method Overview

3 steps: unify diverse data → 3D align in training → same arch at inference



① **Data** 6 public datasets, unified EEF frame and action representation

② **Training** Multi-layer VLA visual tokens ↔ frozen OmniVGGT features

③ **Deployment** Teacher / projector / gates all discarded; zero extra cost

Method Details

Only one training-time auxiliary loss: L = L_action + α · L_align

① Implicit Spatial Alignment

$y = R[\, f^{3D}(I) + E \,]$

$L_align = 1/N \cdot \Sigma\, [1 - \cos(z_i, y_i)]$

Teacher: frozen OmniVGGT, taking RGB plus available camera intrinsics/extrinsics / depth (placeholder tokens if missing)

Target features get positional encodings and are bilinearly resampled to the VLA visual-token resolution

② Adaptive Layer Selection

$z = \Sigma \ell\, g\ell \odot MLP(LN(x\ell))$

$g\ell = \sigma(W_g \cdot x\ell + b_g)$

Inspired by Gated Attention: per-token, per-channel, feature-dependent gating

With heterogeneous data, no single fixed layer suits all domains; LN/MLP and gate parameters are shared across layers

③ State Dropout

p = 25% (best)

Drops the entire proprioceptive-state token per sample independently, keeping images and instructions

Suppresses the "state → demo trajectory" shortcut, forcing the policy to rely on vision

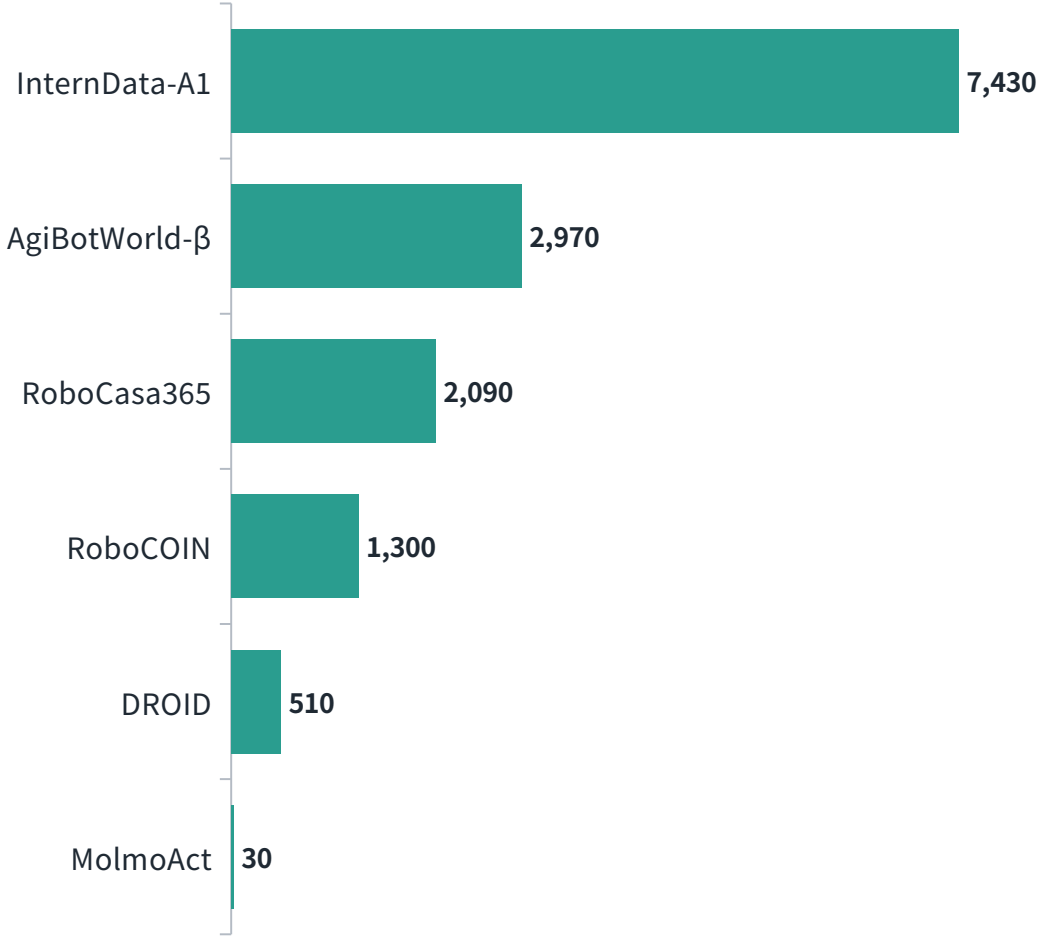
Used in both training stages; disabled at inference

At inference: teacher, projection MLP and gates are all discarded → architecture identical to π0.5, pre-trained weights directly reusable; no depth maps or camera calibration needed as input.

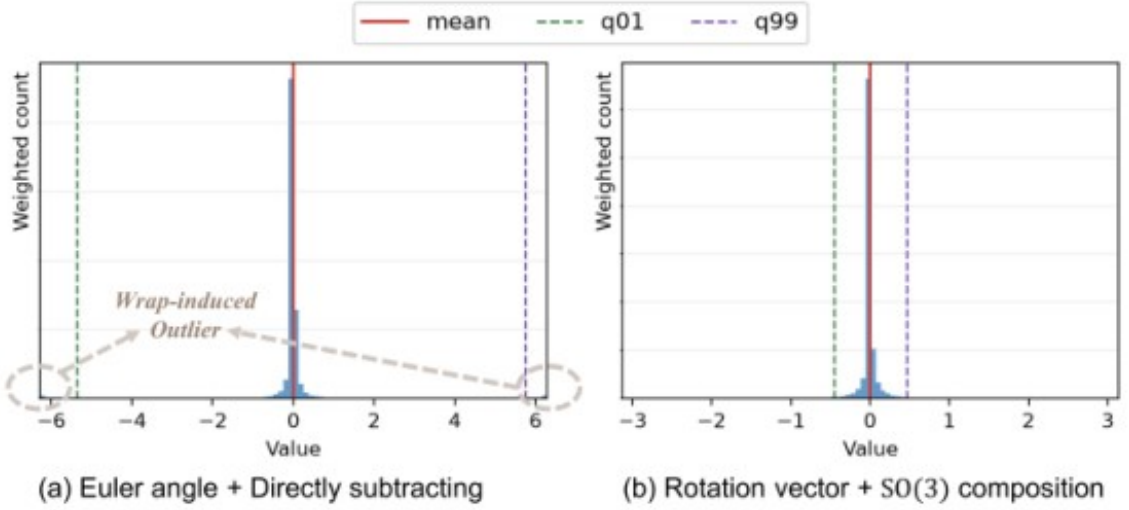
Pre-training Data

1.7M trajectories, 14,330 h: unify frames & actions, then train

Duration per data source (hours)



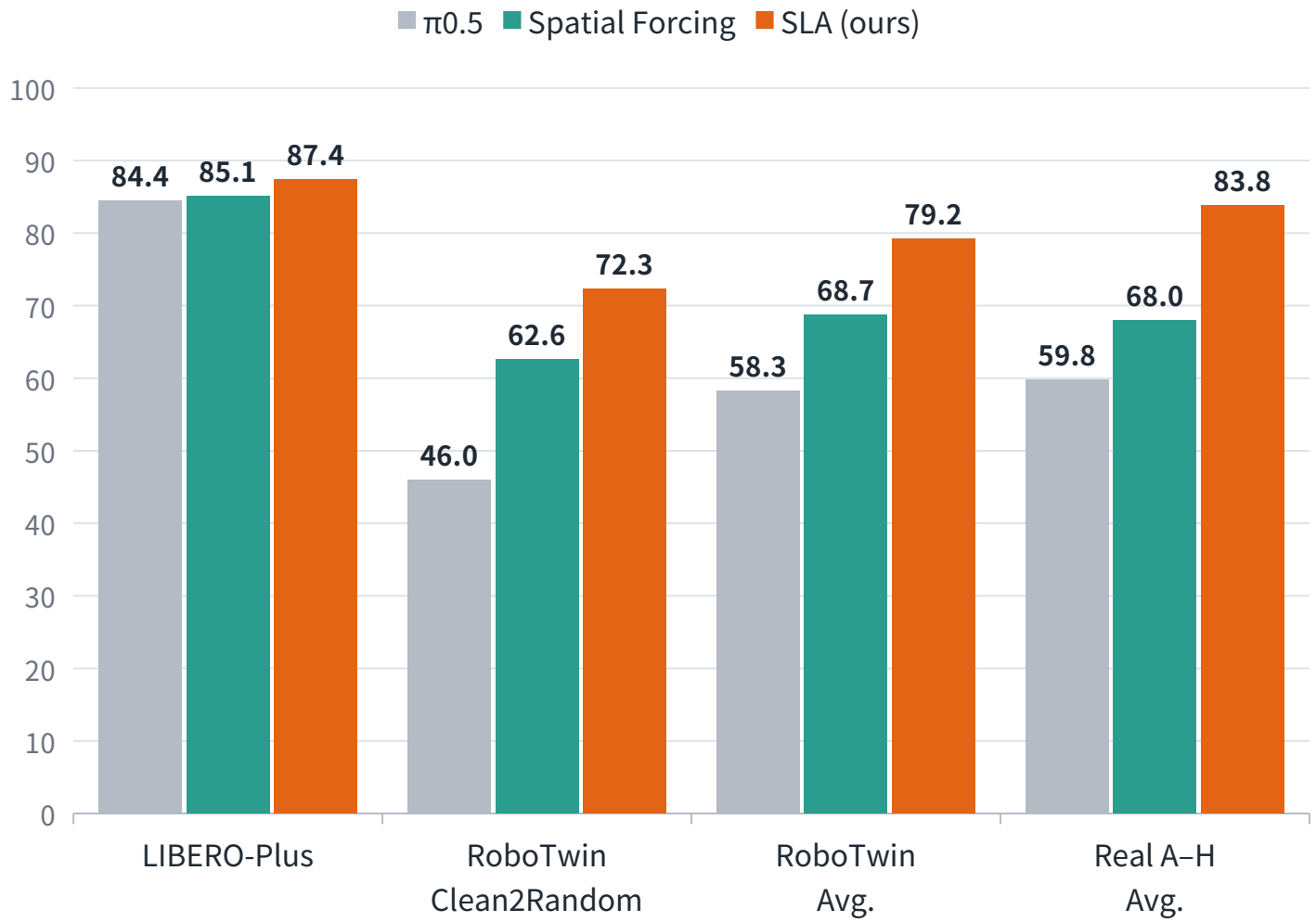
- **Unified frame:** EEF pose, origin at arm base (midpoint for bimanual), axes forward +x / left +y / up +z
- **Actions:** absolute EEF state + in-chunk deltas relative to current pose
- **Downsampling:** only 3–10% of InternData and 0.5% of RoboCasa MimicGen kept
- **Filtering:** remove dexterous hands, base motion, abnormal velocities and static frames; exclude downstream eval data



(a) Euler diff: ±2π wrap outliers (b) Rot. vec + SO(3) comp.: none

Main Results

Beats π0.5 everywhere; biggest gains in hardest generalization settings



RoboCasa-Panda： 68.4%

Best with only 50 demos per task; Cosmos Policy 67.1% (also 50), other methods use 300–3000

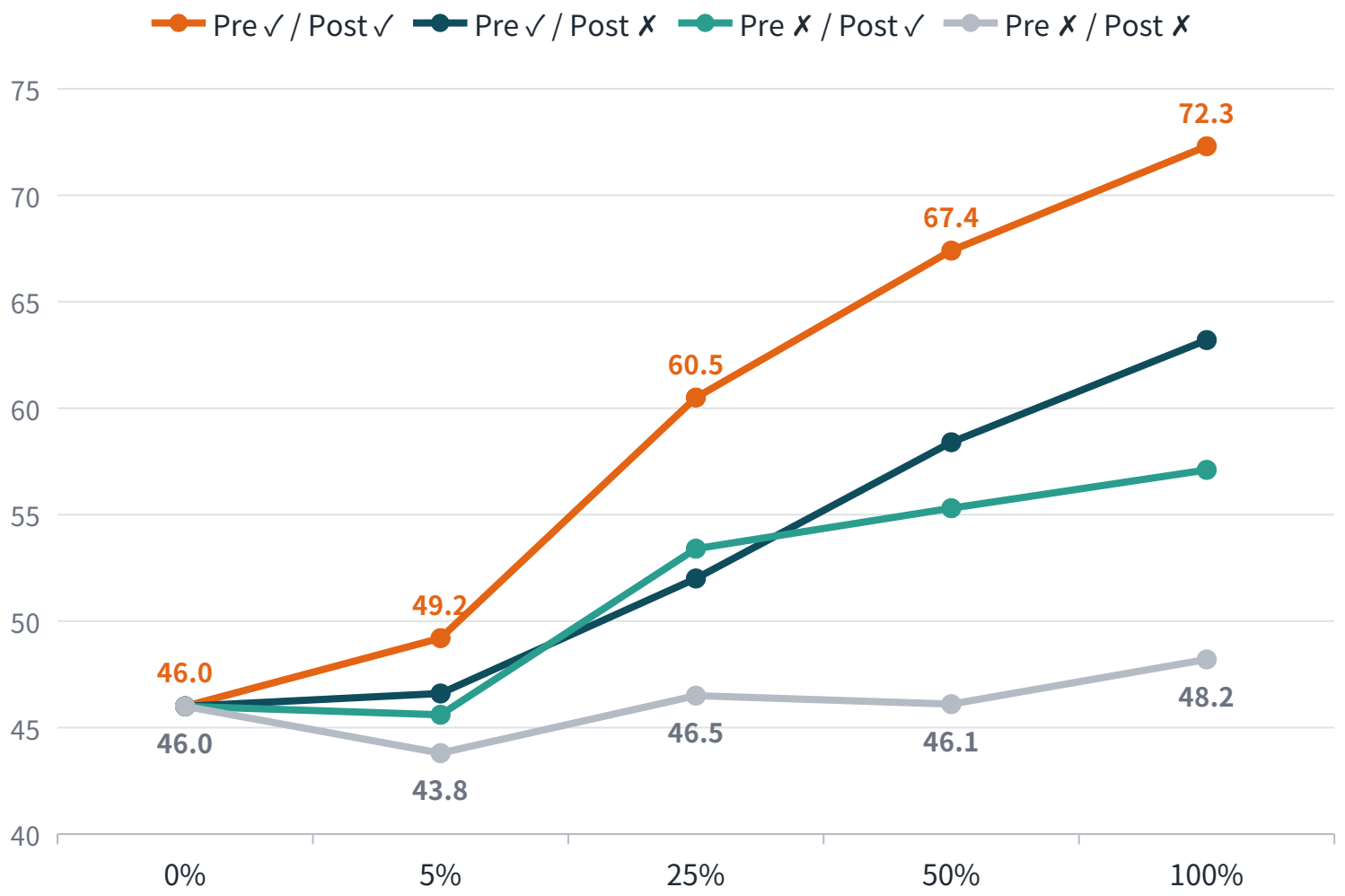
RoboDojo-Real： 33.9% SR

One policy across 3 embodiments; π0.5 12.8%, runner-up InternW0-Δ 31.7% (adds a memory mechanism in training)

Caveat: on LIBERO-Plus and RoboTwin avg., leads runner-up by only 0.3 pts; RoboTwin Clean2Clean 86.0 is below ME-Dex-1.0’ s 89.6

Key Finding: Data Scaling

Data alone barely helps; spatial align makes gains scale



+26.3

Two-stage alignment, 100% data, vs. π0.5 (46.0 → 72.3)

+2.2

No alignment, data only, vs. π0.5 (46.0 → 48.2)

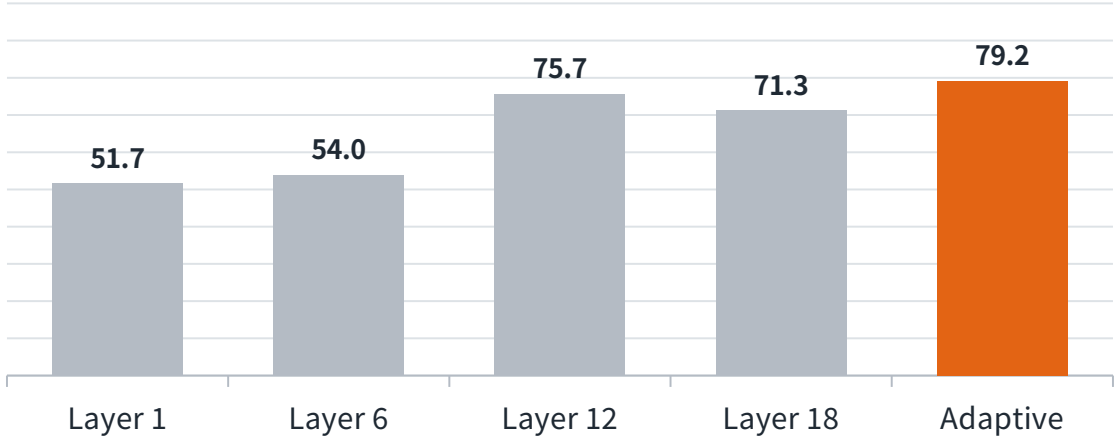
+9.1 / +15.2

Both stages combined vs. pre-training-only / post-training-only alignment

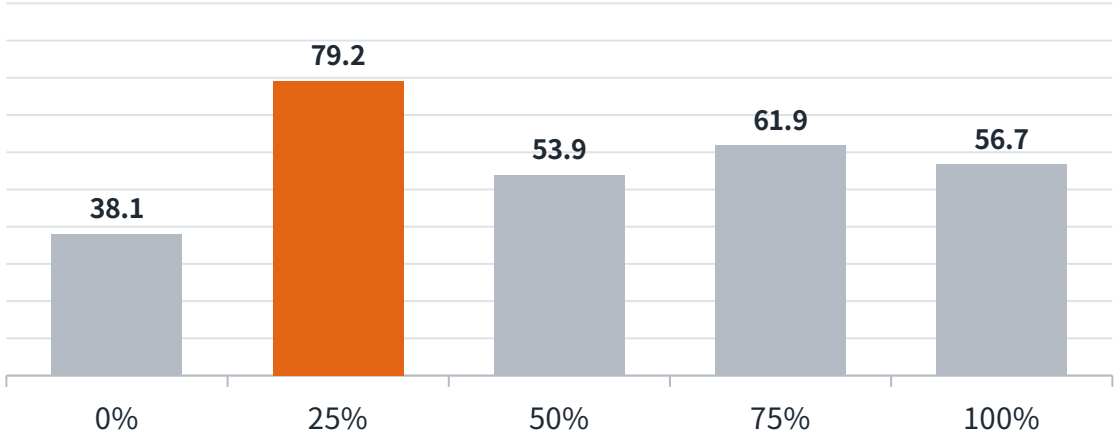
Ablations

Every component helps; state dropout is the most sensitive

Alignment layer selection

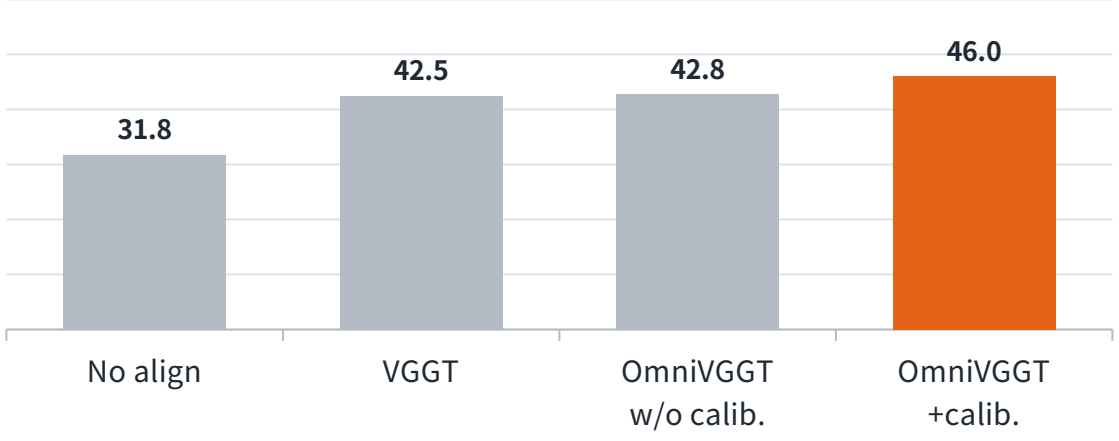


State dropout probability



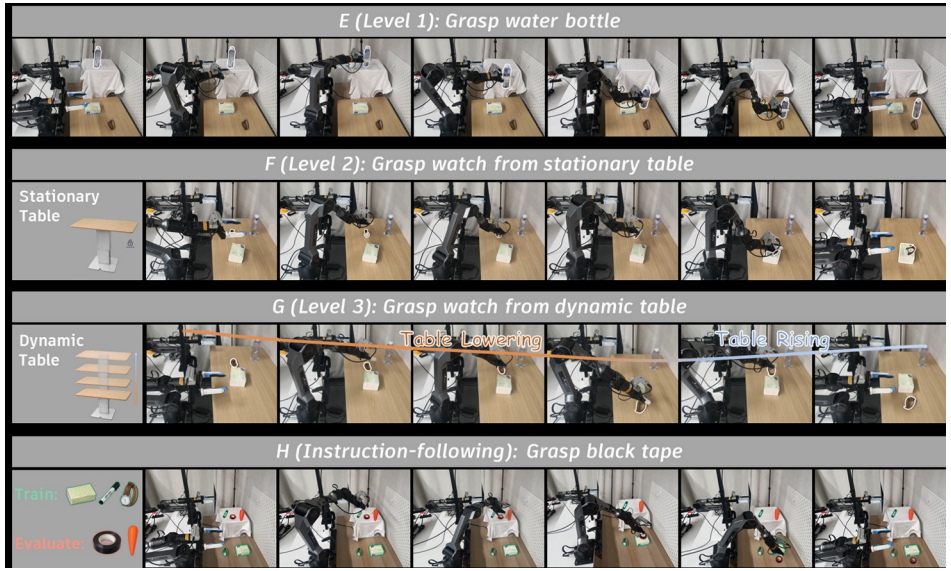
Rotation repr. (all with SO(3) composition)

Alignment teacher / camera calibration



Real-Robot Deployment

First in all 8 real-robot conditions: 83.8% avg. vs. 59.8% for π0.5



A Bimanual handover of 5 vegetables into a bowl (training scene)

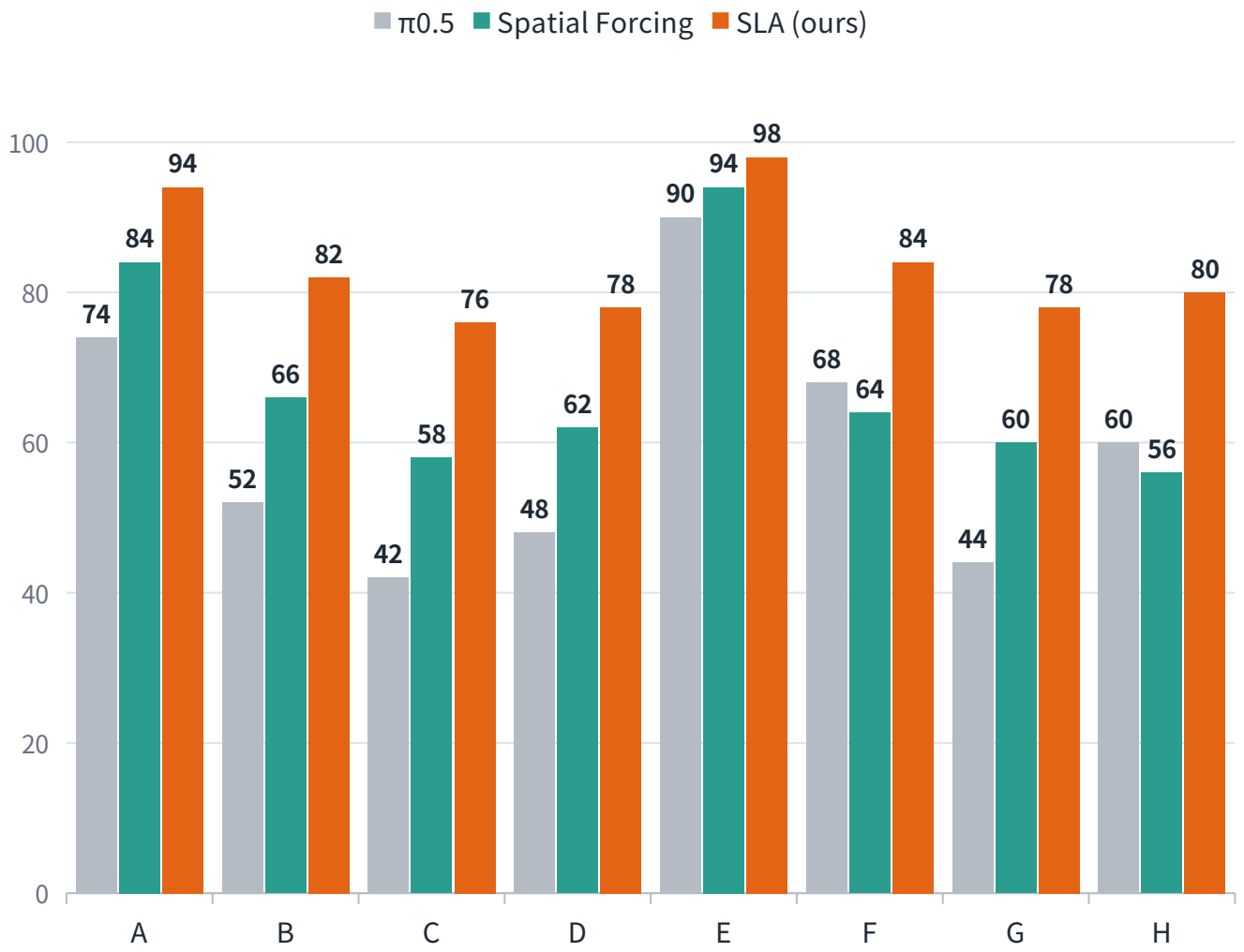
B Human tosses fruit away; robot retrieves it

C Human moves target during grasping

D Cross-embodiment on Franka with only 10 demos

E Grasp water bottle (training scene) **F** Changed table height

G Table raised/lowered during grasp **H** Unseen objects + distractors



Assessment

Practical, well-controlled; but HP-sensitive and some margins tiny

Strengths

- **Zero inference cost:** 3D ability injected only via a training loss; deploys as a drop-in for π0.5 weights
- **Clean controls:** 2×2 alignment settings × 4 data scales show spatial supervision is what makes data scaling work
- **Uses incomplete geometry labels:** OmniVGGT accepts any mix of calibration / depth, so all heterogeneous data is usable
- **Practical engineering recipe:** unified EEF frame, SO(3) composition + rotation vector; large ablation gap (79.2 vs 65.5)

Limitations & Questions

- **Hyperparameter-sensitive:** dropout 25% → 50% drops avg. 79.2 → 53.9; no dropout gives only 38.1, below π0.5's 58.3
- **Narrow ablations:** all on RoboTwin only; best config is the main result
- **Margins:** LIBERO-Plus / RoboTwin avg. only +0.3; RoboDojo-Real uses an extra memory mechanism
- **Evidence strength:** 50 instances per real-robot condition, no variance; depth probe is qualitative only; gate weights not analyzed
- **Cost:** continual pre-training ≈ 48 H100s × 110 hours

Conclusions & Takeaways

Spatial supervision lets VLA CPT absorb data

01

Implicit, No Arch Change

A frozen 3D teacher supervises only during training; deployment is identical to π0.5, with no depth or calibration input

02

Align in Both Stages

Pre- + post-training alignment is best at every data scale; more data alone barely helps (48.2 vs 46.0)

03

Details Set the Ceiling

Multi-layer gating, 25% state dropout, and rotation vector + SO(3) composition each yield clear gains